Frequency domain multiplexing: A method of combining signals for simultaneous transmis-sion by shifting them to different parts of the frequency spectrum.

Frequency dynamic range: The ratio of the largest to the lowest frequency a system can deal with. Analog systems usually have a much larger frequency dynamic range than digital systems.

Frequency resolution: The ability to distinguish or separate closely spaced frequencies.

Frequency response: The magnitude and phase changes that sinusoids experience when passing through a linear system. Usually expressed as a function of frequency. Often found by taking the Fourier transform of the impulse response.

Fricative: Human speech sound that originates as random noise from air turbulence, such as: *s, f, sh, z, v* and *th*. See *voiced* for comparison.

Full-width-at-half-maximum (FWHM): A common way of measuring the width of a peak in a signal. The width of the peak is measured at one-half of the peak's maximum amplitude.

Fundamental frequency: The frequency that a periodic waveform repeats itself. See *harmonic* for comparison.

Gamma curve: The mathematical function or look-up table relating a stored pixel value and the brightness it appears in a displayed image. Also called a *grayscale transform*.

Gaussian: A bell shaped curve of the general form: e^{x^2}. The Gaussian has many unique properties. Also called the *normal distribution*.

Gibbs effect: When a signal is truncated in one domain, ringing and overshoot appear at edges and corners in the other domain.

GIF: A common image file format using LZW (lossless) compression. Widely used on the world wide web for graphics. See *TIFF* and *JPEG* for comparison.

Grayscale: image A digital image where each pixel is displayed in shades of gray between black and white; also called a black and white image.

Grayscale stretch: Greatly increasing the contrast of a digital image to allow the detailed examination of a small range of quantization levels. Quantization levels outside of this range are displayed as saturated black or white.

Grayscale transform: The conversion function between a stored pixel value and the brightness that appears in a displayed image. Also called a *gamma curve*.

Halftone: A common method of printing images on paper. Shades of gray are created by various patterns of small black dots. Color halftones use dots of red, green and blue.

Hamming window: A smooth curve used in the design of filters and spectral analysis, calculated from: $0.54 - 0.46\cos(2\pi n/M)$, where n runs from 0 to M.

Harmonics: The frequency components of a periodic signal, always consisting of integer multiples of the fundamental frequency. The fundamental is the first harmonic, twice this frequency is the second harmonic, etc.

Harvard Architecture: Internal computer layout where the program and data reside in separate memories accessed through separate busses; common in microprocessors used for DSP. See *Von Neumann Architecture* for comparison.

High fidelity: High quality music reproduction, such as provided by CD players.

High-level language: Programming languages such as C, BASIC and FORTRAN.

High-speed convolution: Another name for FFT convolution.

Hilbert transformer: A system having the frequency response: Mag = 1, Phase = 90°, for all frequencies. Used in communications systems for modulation. Can be analog or digital.

Histogram equalization: Processing an image by using the integrated histogram of the image as the grayscale transform. Works by giving large areas of the image higher contrast than the small areas.

Histogram: Displays the distribution of values in a signal. The x-axis show the possible values the samples can take on; the y-axis indicates the number of samples having each value.

Homogeneity: A mathematical property of all linear systems. If an input x[n] produces an output of y[n], then an input $kx[n]$ produces an output of ky[n], for any constant k.

Homomorphic: DSP technique for separating signals combined in a nonlinear way, such as by multiplication or convolution. The nonlinear